

ET 8414 DIGITAL IMAGE PROCESSING

ASSIGNMENT ONE

TO BE SUBMITTED ON 12 MAY 2025

1. Consider the two images' subsets, S1 and S2, shown in the following figure. For $V = \{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent.

	S1	S2
0	0 0 0 0 0	0 0 1 1 0
1	0 0 1 0	0 1 0 0 1
1	0 0 1 0	1 1 0 0 0
0	0 1 1 1	0 0 0 0 0
0	0 1 1 1	0 0 1 1 1

2. Consider the image segment shown.

(a) Let $V = \{0, 1\}$ and compute the lengths of the shortest 4-, 8-, and m-path between p and q. If a particular path does not exist between these two points, explain why.

(b) Repeat for $V = \{1, 2\}$.

3	1	2	1 (q)
2	2	0	2
1	2	1	1
(p)1	0	1	2

3. Implement the noise reduction for the noisy image as shown in Fig. 2.40 (a) (page 112 of the Lecture 1, PPT) and submit your code and the denoised image.

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ASSIGNMENT TWO

TO BE SUBMITTED ON 16 MAY 2025

1. Implement the histogram equalization to the input images 1 and 2; submit your code and the output images.
2. Using Image 3, Reduce the salt-and-pepper noise; submit your code and the output image.
3. Remove the noise from the input images 4,5,6 and 7. Submit your code and the output images.
4. Remove the noise from the input Image 8. Submit your code and the output image.
5. Extract the gradient parts from the input Image 9.
6. Extract the rice objects from the input Image 10.
7. Separate the two types of blobs in the input Image 11.